

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Evaluating generative models requires quantitative evidence that tracks human preference, yet common automatic metrics often provide only partial proxies for perceived quality. Human pairwise comparisons provide a stronger empirical signal, but they are costly, sparse, and noisy because raters vary in expertise, attention, and diligence. Aggregating these measurements therefore requires a model that separates item quality from rater reliability while preserving clear comparisons across methods, datasets, and metrics. Bradley-Terry-based approaches naturally score items from pairwise outcomes, but existing variants either treat all raters as equally reliable or rely on optimization procedures without convergence guarantees, limiting the interpretability of reported margins and uncertainty. We introduce BBQ, a Bayesian Bradley-Terry variant that explicitly estimates rater quality, downweights or removes unreliable participants, and guarantees monotonic likelihood improvement through an Expectation-Maximization algorithm. Across eight human-evaluation dataset conditions, BBQ is best or tied-best for Top-1 agreement throughout the comparison table, achieves the highest Kendall’s Tau on five datasets and second-best values on the remaining three, and remains computationally practical for repeated bootstrap evaluation. The empirical pattern is strongest under noisy or crowdsourced evaluation, where BBQ preserves more of the useful comparison signal while reducing the influence of unreliable raters.

1 INTRODUCTION

Evaluating generative models, including large language models (LLMs) and image generators, requires evidence that reflects human preferences rather than only proxy measurements. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, provide useful numerical summaries but often explain only part of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluation therefore remains the central empirical basis for ranking models when preference quality is the target. This is why leaderboards and comparative studies for large language models and Learned Image Compression (LIC) methods place substantial weight on human preference data.

Human evaluations are expensive, so each collected judgment should contribute as much reliable evidence as possible. Pairwise comparisons, in which participants choose between two items rather than assign absolute scores, are a practical design because they lower the judgment burden and typically produce more consistent observations (Zerman et al., 2018; Wang et al., 2023). However, exhaustive pairwise evaluation is infeasible because the number of pairs grows quadratically with the number of items. Under sparse sampling, raw win rates are not sufficient evidence for global quality: a win rate depends strongly on the quality of the opponents against which it was obtained.

To support leaderboards, pairwise outcomes must be aggregated into comparable item scores with clear ranking margins. The Bradley-Terry (BT) (Bradley & Terry, 1952) model provides this comparison-aware aggregation by updating item scores according to head-to-head outcomes. Because it converts local pairwise evidence into a global probabilistic ranking, BT has become a standard tool for aggregating human judgments across domains.

The quality of the resulting ranking depends both on the estimator and on the raters who generated the comparisons. BT parameters can be estimated with iterative scaling methods (Ford Jr, 1957),

gradient descent, or the minorization maximization (MM) algorithm (Hunter, 2004). For the basic BT model, MM guarantees convergence to the global maximum, whereas gradient-based methods, including approaches closely related to Elo ratings, are widely used but do not provide the same guarantee (Hunter, 2004). For extended BT models that include home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). At the data level, participants may rush, guess, or lose focus, which makes rater reliability heterogeneous. If this heterogeneity is ignored, low-quality raters can shift item scores, reduce stability, and obscure meaningful empirical margins.

We propose a Bayesian Bradley–Terry variant that jointly estimates item quality and rater reliability. To the best of our knowledge, no prior work combines the BT framework with a Bayesian rater-quality model and closed-form EM updates. Existing approaches that account for noisy raters typically use gradient-based optimization without convergence guarantees, whereas our method provides stable monotonic likelihood improvement and interpretable parameters. The Bayesian framework regularizes item scores, represents epistemic uncertainty, and allows rater-specific parameters to quantify how much each participant’s comparisons should influence the final ranking. Through a latent-variable EM formulation, BBQ lets partially reliable raters contribute useful evidence while reducing the impact of inattentive or inconsistent participants. The same model also yields uncertainty estimates for item scores, making ranking comparisons easier to interpret across studies.

We evaluate BBQ on multiple human-evaluation datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. The results are framed around direct comparisons with Bayes-BT and Crowd-BT, using Top-1 agreement, Kendall’s Tau, rater-quality alignment, timing, and uncertainty calibration. Across these empirical settings, BBQ improves robustness to noisy comparisons, remains competitive when raters are already high quality, and converges quickly enough for repeated bootstrapping. Beyond generative model evaluation, the framework applies whenever noisy pairwise comparisons must be transformed into reliable global rankings. Overall, BBQ strengthens the quantitative basis for cost-efficient, interpretable, and reproducible human studies of AI-generated content.

2 RELATED WORK

Human studies often turn many local choices into one global ranking. Presenting participants with a two-item choice rather than a single-item score increases sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). Consequently, many machine-learning leaderboards use pairwise comparisons as their primary empirical evidence. In these setups, participants provide binary preferences, and the aggregation model determines how those local observations become item-level scores and rankings. Notable examples include the Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), which use pairwise comparisons and combine them using variants of the Bradley-Terry (BT) model.

The Bradley-Terry model (Bradley & Terry, 1952) was originally developed to rank competitors in sports, but its comparison structure is broadly useful. It estimates the probability that one item is preferred over another from their relative skill parameters. Thus, a BT ranking is not merely a count of wins; it is adjusted by the inferred strength of the items being compared. Several extensions have been proposed to broaden its applicability. For instance, the Plackett-Luce model generalizes the BT framework from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations by multiplying successive BT probabilities (Luce et al., 1959). Other modifications address specific contexts, such as modeling home-field advantages (Agresti, 2010), incorporating ties (Rao & Kupper, 1967), or handling comparisons between groups instead of individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) is the standard way to convert BT comparison data into parameters. Zermelo (1929) introduced an iterative approach to compute these estimates, which has become widely adopted. Later, Lange et al. (2000) demonstrated that this procedure is a particular case of minorization-maximization (MM) algorithms, which iteratively optimize surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM algorithms

¹The IHQ dataset will be released upon acceptance.

to generalized BT models and established conditions guaranteeing convergence to the MLE. For the classical BT model, the optimization is convex, ensuring convergence to the global optimum (Hunter, 2004). However, for extensions such as home-field advantage, multiple comparisons, or ties, MM algorithms may converge only to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model add regularization and uncertainty estimates to the same comparison evidence (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference via Gibbs sampling. This formulation provides tractable complete-data likelihoods and ensures convergence of the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior estimates, reducing susceptibility to local maxima, and allows for uncertainty quantification in the estimated item skills.

Annotator-quality models address a major source of variation in human-preference evidence: not every comparison is equally reliable. In crowdsourced evaluations, participant quality can vary widely, motivating approaches that explicitly account for annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates both item quality and rater reliability using annotator-specific parameters. Their approach places Gaussian priors on item and rater parameters, and incorporates scaling factors in the likelihood to modulate individual annotator influence. They perform posterior inference using gradient-based optimization, which can be challenging in high-dimensional spaces due to potential local optima. Moreover, gradient-based methods have no guarantees of convergence. By contrast, our EM-based method guarantees monotonic likelihood improvement, making the reported comparison structure more stable. The Elo-based rater model developed by Google Research implements a simplified version of this approach. It is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no prior work combines the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates. In contrast to Chen et al. (2013), who use Gaussian priors and gradient-based optimization without convergence guarantees, our approach leverages conjugate priors and an EM algorithm that ensures monotonic likelihood improvement. Compared to the Elo-based rater model widely used in CLIC (CLIC, 2025), which is a simplified heuristic relying on Elo-style updates, our method provides a principled Bayesian treatment with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

The modeling objective is to convert noisy pairwise comparisons into item scores whose ordering and margins summarize relative quality. Consider a set of K items that are repeatedly compared with one another in pairs by a set of R raters. The data, which we denote as D , consists of the outcomes of these comparisons. For two items i and j of this set, Bradley & Terry (1952) suggested the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating. This probability makes each comparison depend on both competitors, so a win contributes evidence in relation to the strength of the opponent. However, the standard model assumes that all comparisons are equally reliable. Human-evaluation data often violate this assumption because participants differ in expertise, attention, and diligence, giving their judgments different evidential value.

To capture rater heterogeneity, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. With probability q_r , rater r makes an informed judgment following the Bradley-Terry model. With probability $1 - q_r$, the rater guesses randomly, as if flipping a fair coin between the two items. The probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . This likelihood has no closed-form maximum likelihood estimator, so the optimal parameters λ and q cannot be derived analytically. Gradient descent could be used, but it requires optimizer choices and provides no convergence guarantee. On the other hand, the Expectation–Maximization (EM) algorithm avoids learning-rate tuning, and guarantees a monotonic increase of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we use a Thurstonian interpretation of the Bradley-Terry model (Diaconis, 1988). In this perspective, a comparison between items i and j is conceptualized as a race, where each item has a random arrival time, Y_i and Y_j , respectively. These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This leads directly to the standard Bradley-Terry probability used to compare two item skills:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables that separate informative BT comparisons from random guesses. First, we define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which denotes whether the c -th comparison of items i and j by rater r follows the Bradley-Terry model. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent variable formulation keeps the BT comparison probability intact while enabling tractable EM updates that account for rater-specific quality.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm estimates MAP values for the item-skill and rater-quality parameters, λ and q , by treating the rater’s quality and the unobserved arrival times from the Thurstonian interpretation as latent variables. The algorithm is guaranteed to converge to a stationary point of the posterior distribution.

First, we specify prior distributions for the parameters. The item skills λ_k are assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is a conjugate prior for the exponential distribution. Each rater’s quality parameter, q_r , is given a Beta prior, $q_r \sim B(\alpha, \beta)$.

The core of the EM algorithm is the iterative maximization of reliability-weighted complete-data evidence, conditioned on the current parameter estimates (λ^* , q^*). The algorithm proceeds by alternating between two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, a function we denote as Q . This function represents the expected value of the log-posterior of all observed and latent variables, given our observed data and the current parameter estimates from the previous iteration. It is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , is further broken down into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization This step updates the model parameters by maximizing the Q function. By leveraging the expected values from the E-step, the M-step transforms the original complex optimization problem into simpler, closed-form updates. The key quantity that is used in both updates is the posterior probability that a given comparison from rater r follows the Bradley-Terry model. This quantity, denoted as γ , represents the weight of a rater's judgment based on how much it aligns with the model's current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . This posterior probability γ represents our confidence that a comparison was meaningful rather than random, given the current parameter estimates. Higher γ values indicate more trustworthy comparisons.

The new estimate for a rater's quality, q_r , is calculated as a weighted average. The numerator sums up the "effective number of wins" for that rater, where each win is weighted by the posterior probability (γ) that it was a meaningful, non-random judgment. This is combined with the hyperparameters from the Beta prior to regularizing the estimate. The denominator normalizes this sum by the total number of comparisons and prior parameters. This update intuitively increases a rater's quality score if their judgments frequently align with the model's predictions. The update is given by:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The new estimate for an item's skill, λ_i , is a ratio that balances two key quantities. The numerator is a sum of the "effective wins" for item i across all raters, where each win is again weighted by the rater's quality (γ). This term essentially represents the total positive evidence for item i . The denominator, on the other hand, accounts for the total comparisons item i was involved in, and acts as a normalizing factor. These terms are also regularized by the Gamma prior hyperparameters. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The derivation of the Expectation Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate the performance of our **Bayesian Bradley-Terry model with rater Quality (BBQ)**, we conduct a series of experiments comparing it against two baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient descent-based BT model that incorporates rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the implementation provided by Google Research (Google, 2025), which was employed both by CLIC (2025) and Ballé et al. (2025). The hyperparameters used in our experiments can be found in Section C.

We evaluate the performance of several Bradley–Terry (BT) variants across datasets, including human preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), as well as a newly collected inhomogeneous rater quality (IHQ) dataset from the CLIC2024 image test set (CLIC, 2025). To study the effect of rater quality, we partition the IHQ dataset into two subsets: screened and unscreened. This setup enables analysis of how low-quality, high-quality, and mixed comparisons affect model performance. Details of the IHQ dataset creation are given in Section E, and a summary of all datasets, including comparisons, raters, and items, is provided in Section D and Table 2.

Ground truth rankings are not available for real datasets. We first validated the models on simulated datasets, where all methods recover the same ordering as the number of samples increases. For crowd-sourced datasets, we approximate the ground truth by the ranking achieved on the whole dataset. We validate that this provides a reasonable approximation of ordering for real-world datasets by examining the top-1 item in each ranking. For all datasets except IHQ-unscreened, all three models recover the same top-ranked item. For the IHQ-unscreened dataset, Crowd-BT and Bayes-BT fail to identify the reference image as the highest-quality item, whereas BBQ succeeds.

A reliable aggregation method should reproduce the same ordering if the study is repeated. We approximate this stability using bootstrapping, which provides an estimate of the variability in the rankings. Details on the bootstrapping are described in Section F. Note that the WD and HiFiC studies employed active selection of comparison pairs. HiFiC used a binary search strategy, while WD selected pairs based on maximum information gain. For this reason, the bootstrapping results on these two datasets should be interpreted with caution.

We evaluate performance using two metrics. Top-1 agreement is the fraction of bootstrapped samples that identify the same best item as the full dataset. Kendall’s Tau (τ) measures ordinal correlation between rankings (Kendall, 1938), with higher values indicating stable results. Top-1 agreement is most relevant when the best model matters, while Kendall’s Tau assesses overall ranking stability.

4.1 PERFORMANCE ACROSS DATASETS

We compare the three methods across datasets using Top-1 agreement and Kendall’s Tau to quantify winner recovery and full-ranking stability. Top-1 agreement and Kendall’s Tau are summarized in Table 1. Datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are generally easier. Interestingly, HUMAINE deviates from this trend, highlighting that factors beyond the total number of comparisons, such as rater consistency and diversity, can influence model performance.

Overall, BBQ provides the strongest empirical profile across the table. For Top-1 agreement, BBQ is best or tied-best across the dataset conditions, with the largest practical margins in the noisier IHQ settings. For Kendall’s Tau, it recovers the overall ranking most accurately on five of eight datasets and ranks second on the remaining three, showing that gains in winner recovery are accompanied by strong full-ordering stability.

The three datasets where BBQ ranks second in Kendall’s Tau consist of high quality, homogeneous rater sets. On MT-Bench, all models perfectly recover the top item. The WD dataset, collected by Ballé et al. (2025), includes only five raters, likely the paper’s authors, suggesting exceptionally careful evaluation. In IHQ-screened, raters were explicitly filtered for quality. In such settings where rater quality is consistently high, explicitly modeling rater reliability offers little advantage, and the benefits of BBQ over simpler models are reduced.

The IHQ experiments make the practical significance of rater-quality modeling clearest. When low-quality raters are present, BBQ substantially outperforms Crowd-BT and Bayes-BT, especially on IHQ-all and IHQ-unscreened. All three models achieve their best Top-1 accuracy when restricted to the screened subset, highlighting the importance of rater selection. While Crowd-BT explicitly models rater quality, its performance drops noticeably on the full dataset, likely because crowdsourced raters provide fewer than 40 comparisons each, which increases susceptibility to noisy annotations. In contrast, BBQ maintains strong performance without screening, showing that its Bayesian EM weighting can use partially reliable raters while reducing the influence of noisy annotations.

Table 1: Comparison of BT-based aggregation methods across eight dataset conditions. Top: Top-1 agreement [%], the bootstrap frequency with which a method recovers the full-data winner; bottom: Kendall’s Tau, the bootstrap rank correlation with the full-data ordering. Bold marks the best value and underlining marks the second-best value; BBQ is best or tied-best for Top-1 agreement throughout and achieves the highest Kendall’s Tau on five datasets while ranking second on the other three.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	0.9245	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

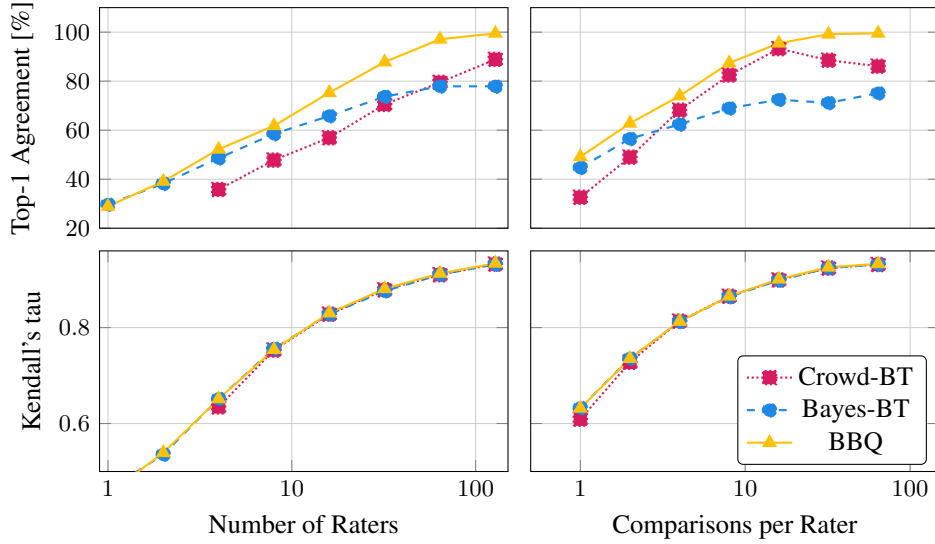


Figure 1: Scaling behavior of Bradley-Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on IHQ-all. **Left:** performance as the number of raters increases with comparisons per rater fixed. **Right:** performance as comparisons per rater increase with the rater pool fixed. Additional evidence improves both Top-1 agreement and Kendall’s τ ; Top-1 agreement shows the clearest between-method separation. Crowd-BT does not converge with very few raters, whereas BBQ remains usable under the EM updates. As rater or comparison counts grow, BBQ increasingly improves over Bayes-BT by estimating and using rater-quality differences.

Nonetheless, achieving uniformly high rater quality is challenging. Large-scale crowdsourcing introduces variability, screening procedures are costly, and subjective factors may affect even diligent annotators. In this context, BBQ provides a principled way to leverage partially reliable raters while reducing the impact of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As Table 1 suggests, the amount of comparison evidence affects both winner recovery and ranking stability. The comparison evidence can increase along two axes: adding more raters, or asking each rater to complete more comparisons.

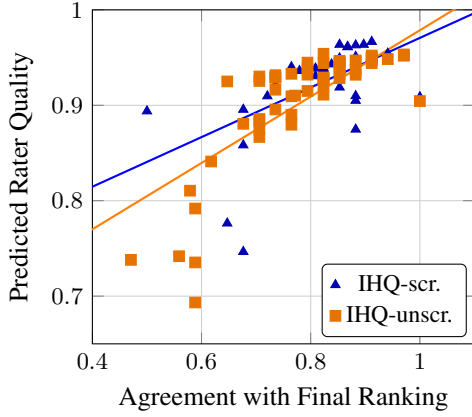


Figure 2: Rater-level validation on IHQ: agreement with the final ranking (x-axis) compared with BBQ predicted rater quality (y-axis). Each point is one rater; triangles show the screened subset and squares show the unscreened subset. The positive correlations show that predicted quality increases with empirical agreement, especially when low-quality raters are included.

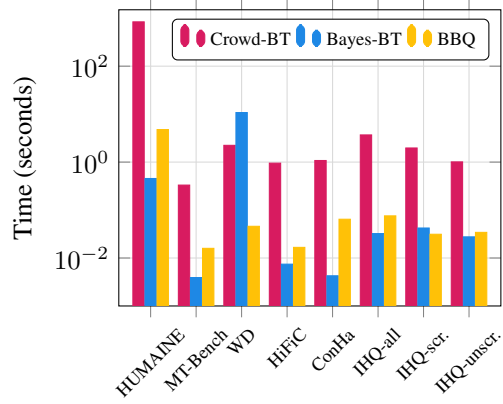


Figure 3: Average computation time in seconds (log-scale) for one bootstrapped sample across eight datasets. The comparison highlights practical runtime: Crowd-BT can require substantially longer optimization on large datasets, while BBQ remains consistently fast across all evaluated conditions.

Figure 1 isolates these two axes for the three BT variants. The left column reports performance as the number of raters increases while comparisons per rater are fixed at the maximum; the right column reports performance as comparisons per rater increase while the rater pool is fixed at the maximum. Top-1 accuracy shows the clearest method-level differences, while Kendall’s τ remains more similar across models. Crowd-BT fails to converge with only one or two raters, highlighting the advantage of using the EM algorithm, which provides convergence guarantees compared to gradient descent.

Crowd-BT and BBQ are similar when the evidence per rater or the number of raters is very small, because there is limited information for estimating individual rater qualities. As the number of raters or comparisons per rater grows, BBQ increasingly outperforms Bayes-BT in Top-1 agreement because it can distinguish rater qualities and weight comparisons accordingly. Bayes-BT can surpass Crowd-BT in the sparsest settings because it avoids estimating rater quality, but this advantage fades once enough data are available for BBQ to use rater heterogeneity productively.

4.3 RATER QUALITY

Figure 2 compares each rater’s agreement with the final ranking against BBQ’s predicted rater quality. We observe a clear positive correlation: raters who agree more closely with the consensus ranking are assigned higher quality by the model. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset, the correlation is stronger, with $r = 0.724$ across 62 raters.

As expected, the unscreened dataset contains several raters with both lower agreement and lower predicted quality. Some raters in the IHQ-unscreened dataset even fall below the level of random guessing (50% agreement), systematically disagreeing with the majority. This pattern quantifies why rater quality matters when aggregating pairwise comparison data. BBQ successfully identifies such raters and assigns them lower quality scores, thereby reducing their influence on the final ranking and mitigating the noise they introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time in seconds required by each method to process a single bootstrapped sample across different datasets. These timings quantify whether the accuracy and stability comparisons are practical for repeated real-world evaluation. BBQ consistently converges

within a few seconds on all datasets. Crowd-BT requires more time than BBQ across the board, with particularly long runtimes on datasets with many comparisons, such as HUMAINE, where convergence takes around 15 minutes. Bayes-BT converges slowly on the WD dataset, though on other datasets it is slightly faster than BBQ. The efficiency of BBQ stems from the closed-form EM updates derived in our method, which enable rapid convergence even on large datasets.

It is also important to note that Crowd-BT was highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ was implemented using plain NumPy without specific optimizations. This suggests that BBQ could be made even faster with a compiled or vectorized implementation. Consequently, BBQ is not only more robust and stable but also highly practical for large-scale human preference studies or applications that require repeated bootstrapping. The observed combination of accuracy, stability, and speed makes BBQ practical for large-scale human preference studies and repeated bootstrapping.

4.5 CONFIDENCE INTERVALS

We evaluate how Crowd-BT, Bayes-BT, and BBQ quantify uncertainty in item rankings. In a controlled simulation, raters provide pairwise comparisons of equally skilled items (H_0). Each model estimates skills and constructs uncertainty intervals to test for significant differences (Section G). The resulting Type I error rates are shown in Figure 6. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ align closely, yielding slightly lower rates, while Bayes-BT is more conservative at $\sim 0.1\%$. These results indicate that BBQ preserves well-calibrated uncertainty estimates while adding rater-quality modeling, whereas Bayes-BT is substantially more conservative.

5 LIMITATIONS

BBQ’s largest empirical gains occur when rater quality is heterogeneous, so its advantage is smaller when all raters are uniformly reliable. This pattern is visible in high-quality or screened settings where simpler baselines are already strong and the margin for rater-quality modeling is limited. Obtaining uniformly reliable raters, however, can require substantial screening cost and may be infeasible in large-scale crowdsourced studies. Thus, BBQ is most valuable in the realistic regime where comparison evidence is useful but uneven in quality.

The current framework is restricted to pairwise comparisons. Many human-evaluation protocols also collect absolute ratings, partial rankings, or multi-way choices, and those formats contain quantitative evidence not modeled directly here. Extending the same Bayesian rater-quality framing to these feedback types would broaden the set of evaluation studies to which the method applies.

Our experiments use bootstrap resampling to quantify ranking stability from the available datasets. This provides a practical measure of variability, but larger independent studies would provide a stronger test of robustness across data-collection protocols. The model also assumes independence across comparisons and does not explicitly model context, presentation order, or fatigue effects, all of which may influence human judgments.

Finally, BBQ is computationally efficient in our experiments, but extremely large numbers of items or raters could still require optimized or compiled implementations. The reported timings suggest that such optimization is a practical engineering path rather than a change to the statistical model.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability from noisy pairwise comparisons. The central contribution is an Expectation-Maximization algorithm that estimates item skills and individual rater-quality parameters with monotonic likelihood improvement. By weighting comparisons according to inferred rater reliability, BBQ reduces the influence of unreliable participants while preserving useful evidence from partially reliable raters. This yields rankings whose scores, uncertainty intervals, and rater-quality estimates are directly interpretable.

486 Across diverse datasets, BBQ delivers the strongest overall quantitative profile among the com-
487 pared BT variants. It is best or tied-best for Top-1 agreement across the reported dataset conditions,
488 achieves the highest Kendall’s Tau on five datasets, and remains second on the remaining three.
489 The largest practical margins appear when noisy or crowdsourced raters are present, especially in
490 IHQ-all and IHQ-unscreened, where BBQ preserves substantially more winner agreement than both
491 Crowd-BT and Bayes-BT. The method also remains fast enough for repeated bootstrapping, and its
492 uncertainty estimates are well calibrated in the controlled Type I error experiment. Predicted rater
493 quality aligns with empirical agreement to final rankings, providing additional validation that the
494 model identifies reliable evaluators.

495 Overall, BBQ strengthens human-evaluation methodology by making the quantitative evidence from
496 noisy pairwise comparisons clearer, more stable, and more actionable. It offers a practical way
497 to aggregate heterogeneous human judgments into reproducible rankings for AI-generated content
498 while reporting the rater reliability and uncertainty needed to interpret those rankings responsibly.
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REFERENCES

- Till Aczel and Roger Wattenhofer. Conditional hallucinations for image compression. *arXiv preprint arXiv:2410.19493*, 2024.
- Eldridge S Adams. Bayesian analysis of linear dominance hierarchies. *Animal Behaviour*, 69(5): 1191–1201, 2005.
- Alan Agresti. *Analysis of ordinal categorical data*. John Wiley & Sons, 2010.
- Jona Ballé, Luca Versari, Emilien Dupont, Hyunjik Kim, and Matthias Bauer. Good, cheap, and fast: Overfitted image compression with wasserstein distortion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 23259–23268, 2025.
- Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Francois Caron and Arnaud Doucet. Efficient bayesian inference for generalized bradley–terry models. *Journal of Computational and Graphical Statistics*, 21(1):174–196, 2012.
- Xi Chen, Paul N Bennett, Kevyn Collins-Thompson, and Eric Horvitz. Pairwise ranking aggregation in a crowdsourced setting. In *Proceedings of the sixth ACM international conference on Web search and data mining*, pp. 193–202, 2013.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael Jordan, Joseph E Gonzalez, et al. Chatbot arena: An open platform for evaluating llms by human preference. In *Forty-first International Conference on Machine Learning*, 2024.
- CLIC. Clic 2024: Challenge on learned image compression. <https://archive.compression.cc/2024/>, 2025. Accessed: 2025-09-12.
- Arthur P Dempster, Nan M Laird, and Donald B Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society: series B (methodological)*, 39(1): 1–22, 1977.
- Persi Diaconis. Group representations in probability and statistics. *Lecture notes-monograph series*, 11:i–192, 1988.
- Lester R Ford Jr. Solution of a ranking problem from binary comparisons. *The American Mathematical Monthly*, 64(8P2):28–33, 1957.
- Rafael C Gonzalez. *Digital image processing*. Pearson education india, 2009.
- Google. Google, 2025. URL <https://github.com/google-research/google-research>. Accessed: 2025-09-12.
- John Guiver and Edward Snelson. Bayesian inference for plackett-luce ranking models. In *proceedings of the 26th annual international conference on machine learning*, pp. 377–384, 2009.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Tzu-Kuo Huang, Ruby C Weng, Chih-Jen Lin, and Greg Ridgeway. Generalized bradley-terry models and multi-class probability estimates. *Journal of Machine Learning Research*, 7(1), 2006.
- David R Hunter. Mm algorithms for generalized bradley-terry models. *The annals of statistics*, 32(1):384–406, 2004.
- Frederick Jelinek. *Statistical methods for speech recognition*. MIT press, 1998.
- Maurice G Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938.
- Brian W. Kernighan and Dennis M. Ritchie. *The C Programming Language*. Prentice Hall, 1988.

594 Kenneth Lange, David R Hunter, and Ilsoon Yang. Optimization transfer using surrogate objective
595 functions. *Journal of computational and graphical statistics*, 9(1):1–20, 2000.

596

597 Mabyduck Ltd. Mabyduck, 2025. URL <https://www.mabyduck.com/>. Accessed: 2025-09-
598 12.

599 R Duncan Luce et al. *Individual choice behavior*, volume 4. Wiley New York, 1959.

600

601 Fabian Mentzer, George D Toderici, Michael Tschannen, and Eiríkur Agustsson. High-fidelity gener-
602 erative image compression. *Advances in neural information processing systems*, 33:11913–11924,
603 2020.

604 Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic
605 evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association
606 for Computational Linguistics*, pp. 311–318, 2002.

607

608 Robin L Plackett. The analysis of permutations. *Journal of the Royal Statistical Society Series C:
609 Applied Statistics*, 24(2):193–202, 1975.

610 Pejaver V Rao and Lawrence L Kupper. Ties in paired-comparison experiments: A generalization
611 of the bradley-terry model. *Journal of the American Statistical Association*, 62(317):194–204,
612 1967.

613

614 Prolific AI Team. Humaine: Human-ai interaction evaluation dataset, 2025. URL [https://
615 huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset](https://huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset).

616 Kexin Wang, Yunlong Zhao, Qianqian Dong, Tom Ko, and Mingxuan Wang. Mospc: Mos prediction
617 based on pairwise comparison. *arXiv preprint arXiv:2306.10493*, 2023.

618

619 Zhou Wang, Eero P Simoncelli, and Alan C Bovik. Multiscale structural similarity for image quality
620 assessment. In *The thirty-seventh asilomar conference on signals, systems & computers, 2003*,
621 volume 2, pp. 1398–1402. Ieee, 2003.

622 Emin Zeman, Vedad Hulusic, Giuseppe Valenzise, Rafał K Mantiuk, and Frédéric Dufaux. The
623 relation between mos and pairwise comparisons and the importance of cross-content comparisons.
624 *Electronic Imaging*, 30:1–6, 2018.

625

626 Ernst Zermelo. Die berechnung der turnier-ergebnisse als ein maximumproblem der wahrschein-
627 lichkeitsrechnung. *Mathematische Zeitschrift*, 29(1):436–460, 1929.

628 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
629 Zi Lin, Zhuohan Li, Dacheng Li, Eric. P Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
630 Judging llm-as-a-judge with mt-bench and chatbot arena, 2023.

A USAGE OF LLMs

During the preparation of this paper, large language models (LLMs) were used only as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, phrasing, and condensation to improve readability. ChatGPT, Claude, and Cursor were also used for code analysis, code completion, and visualization support during development. All core research ideas, experimental design choices, reported comparisons, and interpretation of quantitative results were developed and verified independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items and explicitly track how much evidence each rater contributes. Each rater r has quality q_r : with probability q_r their comparison follows the Bradley–Terry model, and with probability $1 - q_r$ it is random. The following EM (Dempster et al., 1977) derivation shows how these rater-specific reliabilities enter the Bayesian estimator and produce the weighted evidence used in the reported comparisons.

B.1 MODEL DEFINITION

Notation for the item, rater, and comparison-count quantities used in the updates:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley–Terry probability used to compare two item skills:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture probability that combines BT evidence with rater-quality-weighted random choice:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view that supports the EM construction:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

The complete-data objective decomposes the observed comparisons, rater-quality indicators, and latent arrival-time summaries:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-likelihood contribution from the latent arrival-time evidence $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-likelihood contribution from rater-quality indicators $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-likelihood contribution from observed comparison outcomes $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-data log-likelihood after combining the evidence terms:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to (λ, q) while holding the E-step expectations fixed, so the parameter updates are driven by reliability-weighted comparison counts.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit rater-quality update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit item-skill update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters, so the comparison reflects the established baseline configuration.

For Bayes-BT and BBQ, the relevant hyperparameters are the prior distribution parameters and the stopping threshold. Both methods are considered converged when no ELO score changes by more than 1 between two iterations, using the same ELO conversion:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO_SCALE_FACTOR to 400.

We use the same gamma prior for item skill in Bayes-BT and BBQ, with shape 5 and rate 0.1, so item-score regularization is matched across the Bayesian methods. For BBQ, the beta prior on rater quality has $\alpha = 10$ and $\beta = 2$, encoding a prior expectation of mostly reliable raters while leaving room for the data to identify low-quality participants.

D DATASETS

We evaluate the models on human-preference datasets that span language and image domains, providing a range of comparison densities, rater counts, and rater-quality conditions. Table 2 summarizes the quantitative structure of each dataset.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides crowd-worker model comparisons on multi-turn dialogues.

For image compression, we use WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets differ in scale, number of raters, and rater expertise, making them useful for comparing stability under both dense and sparse evidence.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected from a CLIC2024 (CLIC, 2025) user study on the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To quantify the effect of rater quality, we report two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset

Table 2: Dataset summary for the empirical comparisons. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For IHQ, screened subsets include only raters who passed pre-screening tests for attention and display quality, while unscreened subsets include all raters; the columns report the comparison counts and per-rater/model densities used to interpret stability across conditions.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

represents higher-reliability evidence, whereas the unscreened subset reflects the noisier conditions typical of large-scale human evaluation.

Together, these datasets allow the empirical comparisons to cover large clean benchmarks and smaller or noisier settings where rater-reliability modeling is most important.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected using the Mabyduck platform (Ltd., 2025). A screenshot of the platform can be seen in Figure 4. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

To study rater quality, we split the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and completed training comparisons choosing the higher-quality image to reinforce evaluation criteria.

Figure 5 shows four example images from the pre-screening process. The first two are standard color blindness tests, where raters must correctly identify the number shown in each pattern. The final two images are shape detection tests designed to evaluate the raters’ sensitivity to low-contrast objects: one features a light gray shape on a white background, and the other a dark gray shape on a black background. These pre-screening tests help ensure that only raters with adequate visual capabilities contribute to the screened subset.

F BOOTSTRAPPING DETAILS

Because BBQ and Crowd-BT explicitly model rater quality, the bootstrap resamples raters rather than individual comparisons. This preserves each rater’s comparison distribution, so stability is measured under repeated draws of participants rather than artificial shuffling of judgments. For all datasets, we perform 10,000 bootstrap resamples of raters. The exception is HUMAINE, where a single Crowd-BT bootstrap iteration takes approximately 15 minutes as discussed in Section 4.4; for that dataset, we use 1,000 resamples to keep the comparison computationally feasible.

G UNCERTAINTY ESTIMATION DETAILS

In addition to ranking items, uncertainty estimates are needed to decide whether observed score differences are statistically meaningful.

Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) quantify uncertainty through the posterior distribution over item skills. Each skill has a Gamma prior that is

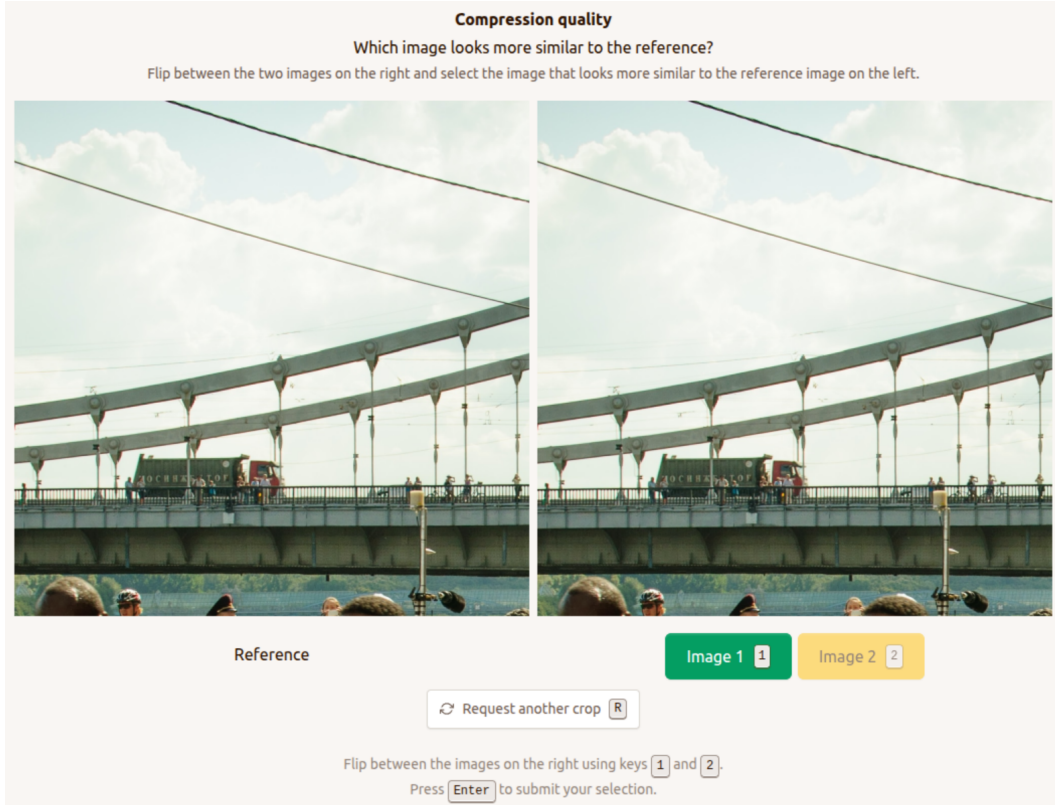


Figure 4: Mabyduck user-study interface used to collect pairwise comparison evidence. The reference image is shown on the left, and the rater chooses between two compressed images on the right, producing the binary preference observations aggregated by the BT-based models.

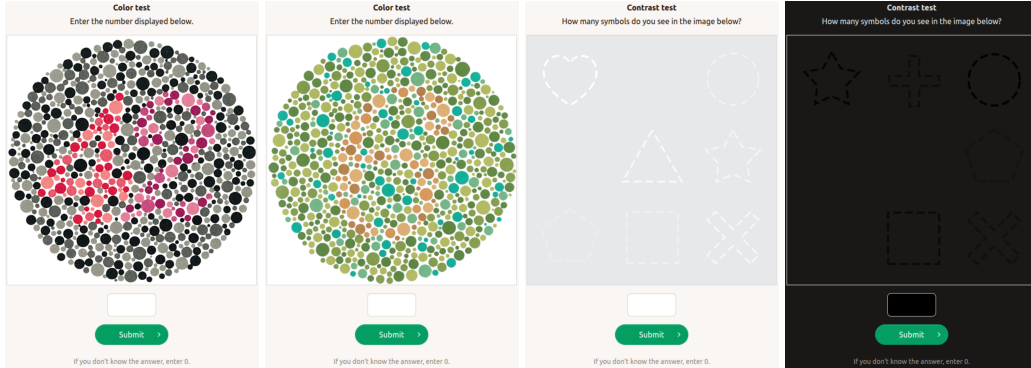


Figure 5: Examples from the rater pre-screening process. The first two images test color perception by asking raters to identify the number in each pattern; the final two test sensitivity to low-contrast objects with light-on-white and dark-on-black shapes.

updated by the observed pairwise comparisons. We convert posterior credible intervals to the Elo scale so uncertainty is reported on the same scale as item scores.

Crowd-BT is non-Bayesian, so we estimate its uncertainty with a frequentist approximation. Specifically, we use a second-order Taylor expansion around the maximum likelihood estimate. The Hessian of the log-likelihood is inverted to obtain the covariance matrix, whose diagonal entries are the item-level variances. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

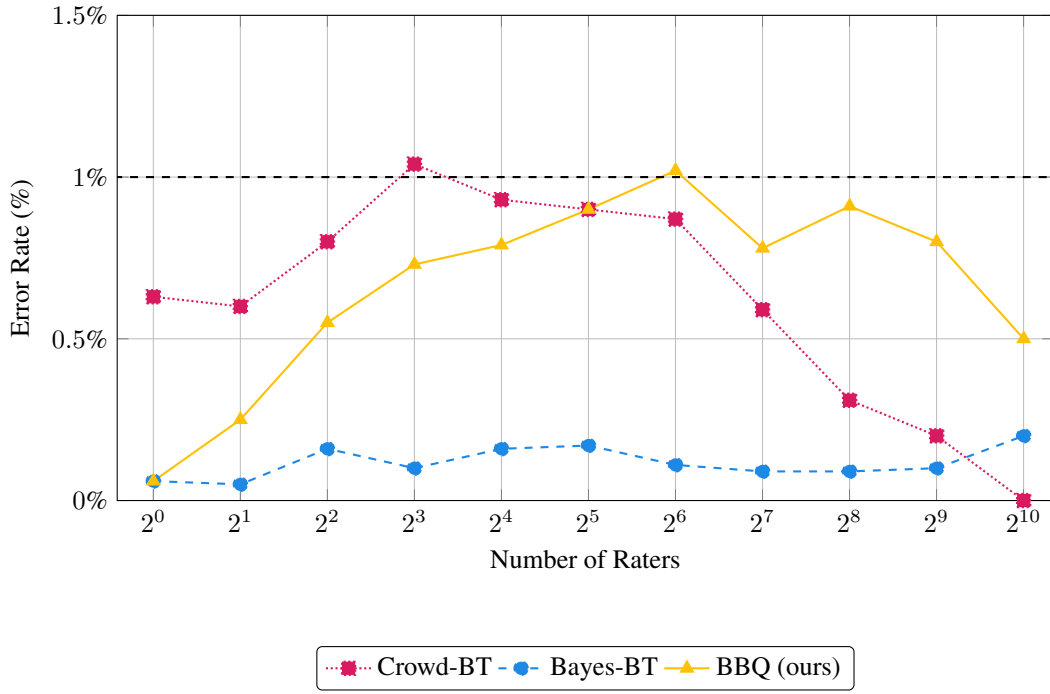


Figure 6: Type I error rate as a function of the number of raters in the equal-skill simulation. The dashed line marks the nominal 1% error rate for 99% intervals, so curves near or below the line indicate calibrated or conservative uncertainty estimates.

where the constants scale the standard deviation to the 99% confidence level. Sharper likelihood peaks produce narrower intervals and therefore higher certainty, while flatter peaks produce wider intervals and greater uncertainty.

We evaluate uncertainty calibration with a controlled simulation experiment. Each trial contains two items with equal skill, corresponding to a 50–50 win probability, and pairwise outcomes are simulated by coin flips. For every trial, we simulate multiple raters and apply each model to estimate Elo scores and uncertainty intervals.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence (or credible) interval for each item’s skill estimate. A model incorrectly rejects H_0 when the intervals do not overlap, because that indicates a significant difference even though the true skills are equal. The primary metric is the observed Type I error rate, i.e., the frequency with which a model falsely reports a difference between equal items.

We perform 10,000 trials with 50 comparisons per rater while varying the number of raters to test how calibration scales with data volume. This setup estimates the empirical Type I error rate for each method and compares it with the theoretical 1% target for 99% intervals. As shown in Figure 6, Crowd-BT and BBQ remain close to the expected 1% level, indicating well-calibrated uncertainty estimates. Bayes-BT is more conservative, consistently producing lower error rates than expected, so its credible intervals are less aligned with nominal significance testing in this simulation.